

INFORMATION THEORY & CODING

Dr. Bo Ning

Full Professor, College of Computer Science

Email: bo.ning@nankai.edu.cn

Review Summary

- **Definition** (*Entropy*) The *entropy* $H(X)$ of a discrete random variable X is defined by

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x).$$

- **Properties of H**

1. $H(X) \geq 0$.
2. $H_b(X) = (\log_b a) H_a(X)$.
3. (*Conditioning reduces entropy*) For any two random variables, X and Y , we have $H(X|Y) \leq H(X)$ with equality **iff** X and Y are independent.
4. (*Independence bound on entropy*)
 $H(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$, with equality **iff** the X_i 's are independent.
5. $H(X) \leq \log |\mathcal{X}|$, with equality **iff** X is distributed uniformly over $\mathcal{X}$.

Review Summary

- **Definition** (*Relative entropy*) The *relative entropy* $D(p||q)$ of the probability mass function p with respect to the probability mass function q is defined by

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

- **Definition** (*Mutual information*) The *mutual information* between two random variables X and Y is defined as

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

Review Summary

■ Properties of D and I

1. $I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X, Y)$.
2. $D(p||q) \geq 0$ with equality iff $p(x) = q(x)$, for all $x \in \mathcal{X}$ (*information inequality*).
3. $I(X; Y) = D(p(x, y)||p(x)p(y)) \geq 0$, with equality iff $p(x, y) = p(x)p(y)$ (i.e., X and Y are independent).
4. If $|\mathcal{X}| = m$, and u is the uniform distribution over $\mathcal{X}$, then $D(p||u) = \log m - H(p)$.

■ Jensen's inequality

If f is a convex function, then $Ef(X) \geq f(EX)$.

Data-processing inequality

If $X \rightarrow Y \rightarrow Z$ forms a Markov chain, then $I(X; Y) \geq I(X; Z)$.

Fano's inequality

- **Problem 2.5** (*Zero conditional entropy*) Show that if $H(X|Y) = 0$, then X is a function of Y , i.e., for all y with $p(y) > 0$, there is **only one** possible value of x with $p(x, y) > 0$.

Proof. Assume that there exists an y , say y_0 and two different values of x , say x_1 and x_2 such that $p(y_0, x_1) > 0$ and $p(y_0, x_2) > 0$. Then $p(y_0) \geq p(y_0, x_1) + p(y_0, x_2) > 0$, and $p(x_1|y_0)$ and $p(x_2|y_0)$ are not equal to 0 or 1. Thus,

$$\begin{aligned} H(X|Y) &= - \sum_y p(y) \sum_x p(x|y) \log p(x|y) \\ &\geq p(y_0) (-p(x_1|y_0) \log p(x_1|y_0) - p(x_2|y_0) \log p(x_2|y_0)) \\ &> 0, \end{aligned}$$

since $-t \log t \geq 0$ for $0 \leq t \leq 1$, and is strictly positive for $t \neq 0, 1$, which is a contradiction to $H(X|Y) = 0$.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y . Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y . Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.
- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y . Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.
- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

Why do we need to related P_e to the entropy $H(X|Y)$?

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y . Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.
- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

Why do we need to related P_e to the entropy $H(X|Y)$?

When we have a communication system, we send X , but receive a corrupted version Y . We want to infer X from Y . Our estimate is $\hat{X}$ and we will make a mistake as

$$P_e = \Pr[\hat{X} \neq X]$$

Markov chain $X \rightarrow Y \rightarrow \hat{X}$.

Fano's inequality

- **Problem.** A random variable Y is related to another random variable X with a distribution $p(x)$. From Y , we calculate a function $g(Y) = \hat{X}$, where $\hat{X}$ is an estimate of X and takes on values in $\hat{\mathcal{X}}$. We observe that $X \rightarrow Y \rightarrow \hat{X}$ forms a Markov chain. How to bound the estimate error probability $P_e = \Pr[\hat{X} \neq X]$?

Fano's inequality

- **Theorem 2.10.1** (*Fano's inequality*) For any estimator $\hat{X}$ such that $X \rightarrow Y \rightarrow \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

This inequality can be weakened to

$$1 + P_e \log(|\mathcal{X}| - 1) \geq H(X|Y)$$

or

$$P_e \geq \frac{H(X|Y) - 1}{\log |\mathcal{X}| - 1}.$$

Proof.

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \rightarrow Y \rightarrow \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

Proof. Define an error random variable as

$$E = \begin{cases} 1 & \text{if } \hat{X} \neq X, \\ 0 & \text{if } \hat{X} = X. \end{cases}$$

Using the chain rule for entropies to expand $H(E, X|\hat{X})$ in two different ways, we have

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}| - 1)}. \end{aligned}$$

Since *conditioning reduces entropy*, $H(E|\hat{X}) \leq H(E) = H(P_e)$. Since E is a function of X and $\hat{X}$, the conditional entropy $H(E|X, \hat{X})$ is equal to 0. We now look at $H(X|E, \hat{X})$. By the equation $H(X|Y) = \sum_y p(y)H(X|Y = y)$, we have

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{ \Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1) \}. \end{aligned}$$

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \rightarrow Y \rightarrow \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

Proof.

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}| - 1)}. \end{aligned}$$

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{ \Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1) \}. \end{aligned}$$

By definition of E , X is **conditionally deterministic** given $\hat{X} = \hat{x}$ and $E = 0$, then $H(X|\hat{X} = \hat{x}, E = 0) = 0$. If $\hat{X} = \hat{x}$ and $E = 1$, then X must take a value in the set $\{x \in \mathcal{X} : x \neq \hat{x}\}$ which contains $|\mathcal{X}| - 1$ elements. Then $H(X|\hat{X} = \hat{x}, E = 1) \leq \log(|\mathcal{X}| - 1)$.

$$\begin{aligned} H(X|E, \hat{X}) &\leq \sum_{\hat{x} \in \mathcal{X}} \Pr[\hat{X} = \hat{x}, E = 1] \log(|\mathcal{X}| - 1) \\ &= \Pr[E = 1] \log(|\mathcal{X}| - 1) \\ &= P_e \log(|\mathcal{X}| - 1). \end{aligned}$$

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \rightarrow Y \rightarrow \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

Proof.

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}| - 1)}. \end{aligned}$$

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{ \Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1) \}. \end{aligned}$$

$$\begin{aligned} H(X|E, \hat{X}) &\leq \sum_{\hat{x} \in \mathcal{X}} \Pr[\hat{X} = \hat{x}, E = 1] \log(|\mathcal{X}| - 1) \\ &= \Pr[E = 1] \log(|\mathcal{X}| - 1) \\ &= P_e \log(|\mathcal{X}| - 1). \end{aligned}$$

By the data-processing inequality, we have $I(X; \hat{X}) \leq I(X; Y)$ and therefore $H(X|\hat{X}) \geq H(X|Y)$.

Corollary

- **Corollary** For any two random variables X and Y , let $p = \Pr(X \neq Y)$.

$$H(p) + p \log(|\mathcal{X}| - 1) \geq H(X|Y).$$

Proof. Let $\hat{X} = Y$ in Fano's inequality.

Fano's inequality

- **Remark** Suppose that there is **no knowledge** of Y . Thus, X must be guessed without any information. Let $X \in \{1, 2, \dots, m\}$ and $p_1 \geq p_2 \geq \dots \geq p_m$. Then the best guess of X is $\hat{X} = 1$ and the resulting probability of error is $P_e = 1 - p_1$. Fano's inequality becomes

$$H(P_e) + P_e \log(m - 1) \geq H(X).$$

The probability mass function

$$(p_1, p_2, \dots, p_m) = (1 - P_e, \frac{P_e}{m - 1}, \dots, \frac{P_e}{m - 1})$$

achieves this bound with equality.

Fano's inequality

由此得到如下的不等式：

引理 2.10.1 如果 X 和 X' 独立同分布，具有熵 $H(X)$ ，则

$$\Pr(X=X') \geq 2^{-H(X)}$$

当且仅当 X 服从均匀分布，等号成立。

证明：假定 $X \sim p(x)$ 。由 Jensen 不等式，可得

$$2^{E \log p(X)} \leq E 2^{\log p(X)}$$

含义是

$$2^{-H(X)} = 2^{\sum p(x) \log p(x)} \leq \sum p(x) 2^{\log p(x)} = \sum p^2(x)$$

推论 设 X 和 X' 相互独立，且 $X \sim p(x)$, $X' \sim r(x)$, $x, x' \in \mathcal{X}$ ，那么

$$\Pr(X=X') \geq 2^{-H(p) - D(p \parallel r)}$$

$$\Pr(X=X') \geq 2^{-H(r) - D(r \parallel p)}$$

证明：我们有

$$\begin{aligned} 2^{-H(p) - D(p \parallel r)} &= 2^{\sum p(x) \log p(x) + \sum p(x) \log \frac{r(x)}{p(x)}} \\ &= 2^{\sum p(x) \log r(x)} \\ &\leq \sum p(x) 2^{\log r(x)} \\ &= \sum p(x) r(x) \\ &= \Pr(X=X') \end{aligned}$$

其中的不等式可由 Jensen 不等式和函数 $f(y) = 2^y$ 得到。

Fano's inequality

- **Lemma 2.10.1** If X and X' are i.i.d. with entropy $H(X)$,

$$\Pr[X = X'] \geq 2^{-H(X)},$$

with equality iff X has a uniform distribution.

- **Corollary** Let X, X' be independent with $X \sim p(x)$, $X' \sim r(x)$, $x, x' \in \mathcal{X}$. Then

$$\Pr[X = X'] \geq 2^{-H(p) - D(p||r)}$$

$$\Pr[X = X'] \geq 2^{-H(r) - D(r||p)}$$

Stock Market

- Initial investment Y_0 , daily return ratio r_i , in the t -th day, your money is

$$Y_t = Y_0 r_1 \cdot \dots \cdot r_t.$$

- Now if returns ratio r_i 's are i.i.d., with

$$r_i = \begin{cases} 4, & w.p. 1/2 \\ 0, & w.p. 1/2 \end{cases}$$

- So you think the expected return ratio is $Er_i = 2$.

- and then

$$EY_t = E(Y_0 r_1 \cdot \dots \cdot r_t) = Y_0 (Er_i)^t = Y_0 2^t ???$$

Stock Market

- With $Y_0 = 1$, actual return Y_t goes like

1 4 16 0 0 ...

- *Why?*: products does not behave the same as addition!

The Markov and Chebyshev Inequalities

- **Theorem** (*Markov Inequality*) Suppose that X is a random variable such that $\Pr[X \geq 0] = 1$. Then for every real number $t > 0$,

$$\Pr[X \geq t] \leq \frac{EX}{t}.$$

Proof.

$$EX = \sum_x xp(x) = \sum_{x < t} xp(x) + \sum_{x \geq t} xp(x).$$

Since X can have only nonnegative values, we have

$$EX \geq \sum_{x \geq t} xp(x) \geq \sum_{x \geq t} tp(x) = t\Pr[X \geq t].$$

The Markov and Chebyshev Inequalities

- **Theorem** (*Chebyshev Inequality*) Let X be a random variable for which $\text{Var}(X)$ exists. Then for every number $t > 0$,

$$\Pr[|X - EX| \geq t] \leq \frac{\text{Var}(X)}{t^2}.$$

Proof. Let $Y = (X - EX)^2$. Then $\Pr[Y \geq 0] = 1$ and $EY = \text{Var}(X)$. By applying the Markov inequality to Y , we obtain:

$$\Pr[|X - EX| \geq t] = \Pr[Y \geq t^2] \leq \frac{\text{Var}(X)}{t^2}.$$

Weak Law of Large Numbers

- **Theorem** (*Weak Law of Large Numbers*) Suppose that $X_1, X_2, \dots, X_n$ are n independent, identically distributed (i.i.d.) random variables, then

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow EX \quad \text{in probability,}$$

i.e., for every number $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \Pr\left[\left|\frac{1}{n} \sum_{i=1}^n X_i - EX\right| \leq \epsilon\right] = 1.$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^n X_i}$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^n X_i}$$

- We can write

$$X_i = e^{\log X_i}$$

Hence

$$\sqrt[n]{\prod_{i=1}^n X_i} = e^{\frac{1}{n} \sum_{i=1}^n \log X_i}$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^n X_i}$$

- We can write

$$X_i = e^{\log X_i}$$

Hence

$$\sqrt[n]{\prod_{i=1}^n X_i} = e^{\frac{1}{n} \sum_{i=1}^n \log X_i}$$

$$\sqrt[n]{\prod_{i=1}^n X_i} \rightarrow e^{E(\log X)} \leq e^{\log EX} = EX.$$

L.L.N for Product of Random Variables

- How does this behave?

$$\sqrt[n]{\prod_{i=1}^n X_i}$$

- We can write

$$X_i = e^{\log X_i}$$

Hence

$$\sqrt[n]{\prod_{i=1}^n X_i} = e^{\frac{1}{n} \sum_{i=1}^n \log X_i}$$

$$\sqrt[n]{\prod_{i=1}^n X_i} \rightarrow e^{E(\log X)} \leq e^{\log EX} = EX.$$

$$\text{geometric mean } \sqrt[n]{\prod_{i=1}^n X_i} \leq \text{arithmetic mean } \frac{1}{n} \sum_{i=1}^n X_i$$

L.L.N for Product of Random Variables

- Stock example:

$$E \log r_i = \frac{1}{2} \log 4 + \frac{1}{2} \log 0 = -\infty$$

$$E(Y_t) \rightarrow Y_0 e^{E \log r_i} = 0, \quad t \rightarrow \infty$$

L.L.N for Product of Random Variables

- Stock example:

$$E \log r_i = \frac{1}{2} \log 4 + \frac{1}{2} \log 0 = -\infty$$

$$E(Y_t) \rightarrow Y_0 e^{E \log r_i} = 0, \quad t \rightarrow \infty$$

- Example:

$$X = \begin{cases} a & \text{w.p. } 1/2 \\ b & \text{w.p. } 1/2 \end{cases}$$

$$E \left(\sqrt[n]{\prod_{i=1}^n X_i} \right) \rightarrow \sqrt{ab} \leq \frac{a+b}{2}$$

Asymptotic Equipartition Property (AEP)

■ **Definition** (*Convergence of random variables*). Given a sequence of random variables, $X_1, X_2, \dots$, we say that the sequence $X_1, X_2, \dots$ *converges* to a random variable X :

1. *In probability* if for every $\epsilon > 0$,
 $\Pr[|X_n - X| \geq \epsilon] \rightarrow 0$
2. *In mean square* if $E(X_n - X)^2 \rightarrow 0$
3. *With probability 1* (a.k.a. *almost surely*) if
 $\Pr[\lim_{n \rightarrow \infty} X_n = X] = 1$

Asymptotic Equipartition Property (AEP)

- **Theorem 3.1.1** (*AEP*) If $X_1, X_1, \dots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X) \quad \text{in probability.}$$

Asymptotic Equipartition Property (AEP)

- **Theorem 3.1.1** (*AEP*) If $X_1, X_1, \dots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X) \quad \text{in probability.}$$

Proof. Since X_i 's are i.i.d., so are $\log p(X_i)$'s. Hence, by the *weak law of large numbers*,

$$\begin{aligned} -\frac{1}{n} \log p(X_1, X_2, \dots, X_n) &= -\frac{1}{n} \sum_i \log p(X_i) \\ &\rightarrow -E \log p(X) \quad \text{in probability} \\ &= H(X), \end{aligned}$$

AEP, Typical Set

- AEP lies in the *heart* of information theory.
 - ◇ Proof for lossless source coding
 - ◇ Proof for channel capacity
 - ◇ and more...

AEP, Typical Set

- AEP lies in the *heart* of information theory.
 - ◇ Proof for lossless source coding
 - ◇ Proof for channel capacity
 - ◇ and more...
- **Definition** A *typical set* $A_\epsilon^{(n)}$ contains all sequences $(x_1, x_2, \dots, x_n) \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \leq p(x_1, x_2, \dots, x_n) \leq 2^{-n(H(X)-\epsilon)}.$$

■ Theorem 3.1.2

1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then
$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A .
4. $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Properties of AEP

■ Theorem 3.1.2

1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then
$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A .
4. $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 1. Immediate from the definition of $A_\epsilon^{(n)}$.

The number of bits used to describe sequences in typical set is approximately $nH(X)$.

Properties of AEP

■ Theorem 3.1.2

1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then
$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A .
4. $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 2. By Theorem 3.1.1, the probability of the event $(X_1, X_2, \dots, X_n) \in A_\epsilon^{(n)}$ tends to 1 as $n \rightarrow \infty$. Thus, for any $\delta > 0$, there exists an n_0 such that for all $n \geq n_0$, we have

$$\Pr\left\{\left| -\frac{1}{n} \log p(X_1, X_2, \dots, X_n) - H(X) \right| < \epsilon\right\} > 1 - \delta.$$

Setting $\delta = \epsilon$, the conclusion follows.

Properties of AEP

■ Theorem 3.1.2

1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then
$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A .
4. $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 3.

$$\begin{aligned} 1 &= \sum_{\mathbf{x} \in \mathcal{X}^n} p(\mathbf{x}) \geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x}) \\ &\geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X)+\epsilon)} \\ &= 2^{-n(H(X)+\epsilon)} |A_\epsilon^{(n)}|. \end{aligned}$$

Properties of AEP

■ Theorem 3.1.2

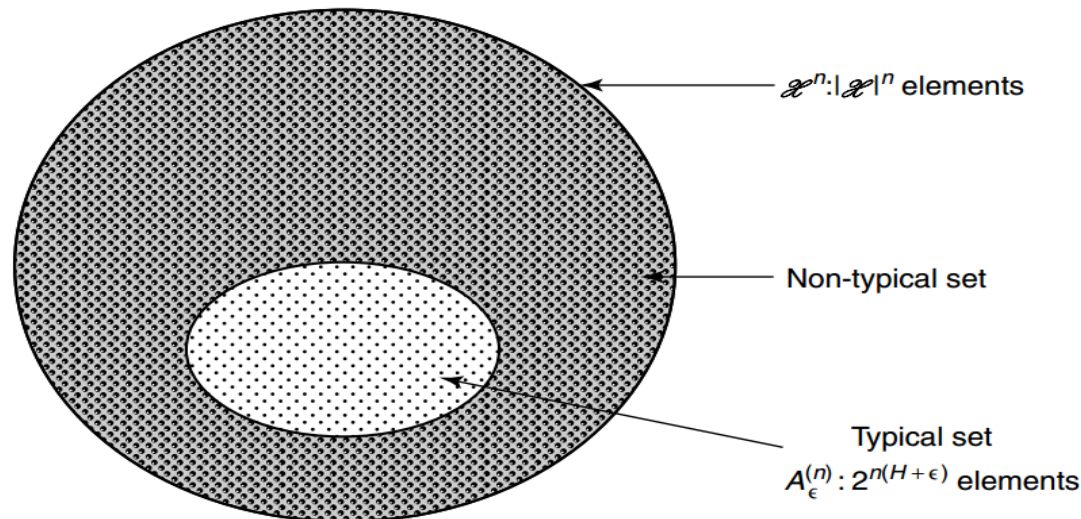
1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then
$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$
2. $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$ for n sufficiently large.
3. $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the cardinality of the set A .
4. $|A_\epsilon^{(n)}| \geq (1 - \epsilon)2^{n(H(X)-\epsilon)}$ for n sufficiently large.

Proof. 4. For sufficiently large n , $\Pr[A_\epsilon^{(n)}] > 1 - \epsilon$, so that

$$\begin{aligned} 1 - \epsilon &< \Pr[A_\epsilon^{(n)}] \\ &\leq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X)-\epsilon)} \\ &= 2^{-n(H(X)-\epsilon)} |A_\epsilon^{(n)}|. \end{aligned}$$

Typical set diagram

- This enables us to divide all sequences into two sets
 - Typical set: high probability to occur, sample entropy is close to true entropy
so we will focus on analyzing sequences in typical set
 - Non-typical set: small probability, can ignore in general



Asymptotic Equipartition Property (AEP)

- **Theorem 3.2.1** Let X^n be i.i.d. $\sim p(x)$. Let $\epsilon > 0$. Then there exists a code that maps sequences x^n of length n into binary strings such that the mapping is one-to-one (and therefore invertible) and

$$E\left[\frac{1}{n}\ell(X^n)\right] \leq H(X) + \epsilon$$

for n sufficiently large.

Asymptotic Equipartition Property (AEP)

■ Theorem 3.2.1

$$E\left[\frac{1}{n}\ell(X^n)\right] \leq H(X) + \epsilon$$

for n sufficiently large.

Proof. Description in *typical set* requires no more than $n(H(X) + \epsilon) + 1$ bits (correction of 1 bit because of integrality).

Description in *atypical set* $A_\epsilon^{(n)c}$ requires no more than $n \log |\mathcal{X}| + 1$ bits.

Add **another bit** to indicate whether in $A_\epsilon^{(n)}$ or not to get whole description.

Asymptotic Equipartition Property (AEP)

■ Theorem 3.2.1

$$E\left[\frac{1}{n}\ell(X^n)\right] \leq H(X) + \epsilon$$

for n sufficiently large.

Proof. (cont.) Let $\ell(x^n)$ be the length of the binary description of x^n . Then, $\forall \epsilon > 0$, there exists n_0 s.t. $\forall n > n_0$,

$$\begin{aligned} E(\ell(X^n)) &= \sum_{x^n} p(x^n) \ell(x^n) \\ &= \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) \ell(x^n) + \sum_{x^n \in A_\epsilon^{(n)C}} p(x^n) \ell(x^n) \\ &\leq \sum_{x^n \in A_\epsilon^{(n)}} p(x^n) (n(H + \epsilon) + 2) \\ &\quad + \sum_{x^n \in A_\epsilon^{(n)C}} p(x^n) (n \log |\mathcal{X}| + 2) \\ &= \Pr[A_\epsilon^{(n)}] (n(H + \epsilon) + 2) + \Pr[A_\epsilon^{(n)C}] (n \log |\mathcal{X}| + 2) \\ &\leq n(H + \epsilon) + \epsilon n (\log |\mathcal{X}|) + 2 \\ &= n(H + \epsilon'), \end{aligned}$$

where $\epsilon' = \epsilon + \epsilon \log |\mathcal{X}| + \frac{2}{n}$ can be made arbitrarily small by choosing n properly.

INFORMATION THEORY & CODING